where the first term can be calculated by $\log p_{\tilde{\phi}}(\mathbf{x}) = \sum_{t=1}^{T} \log p_{\tilde{\phi}}(\mathbf{x}_t|\mathbf{x}_{<t})$.

Putting it altogether, to select the ICL examples, we evaluate $\delta = \log p_{\tilde{\phi}}(\mathbf{x}) - \log p_\theta(\mathbf{x})$ for each statement $\mathbf{x} \in \mathcal{S}_{\tilde{\phi}}$, and select the top-$K$ statements with the highest δ score. Then, the selected examples are prepended in the same format as the original query, as exemplified in Figure 1. In testing, our mapping function predicts the action of agreeing or disagreeing with the given test statement $\mathbf{x}_{\text{test}}$, defined as:

$$f_\phi(\mathbf{x}_{\text{test}}) = \arg\max_{a \in \mathcal{A}} p_\theta(a|[(\mathbf{x}_i, a_i)]_{i \in \mathcal{I}_{\tilde{\phi}}}, \mathbf{x}_{\text{test}}),$$

$$\text{where} \quad \mathcal{I}_{\tilde{\phi}} = \arg\max_{I \subset \{1,\cdots,|\mathcal{S}_{\tilde{\phi}}|\}; |I|=K} \sum_{i \in I} \delta_i,$$

which is the set of indices for the examples selected from $\mathcal{S}_{\tilde{\phi}}$. We summarize our end-to-end framework in Algorithm 1.

3.2. Evaluating Persona Elicitation

In this subsection, we systematically define four evaluation metrics for the general Persona Elicitation task: **(1)** Action Consistency, **(2)** Action Confidence, **(3)** Action Uncertainty, and **(4)** Degree of Alteration. Each metric is computed over a set of test input statements $\mathcal{T}_{\tilde{\phi}} = \{\mathbf{x}_i\}_{i=1}^m$.

Definition 1. ***(Action Consistency)*** *is defined as the fraction of predicted actions $\hat{a}_i = f_\phi(\mathbf{x}_i)$ matching ground truth a_i:*

$$\frac{1}{m} \sum_{i=1}^{m} \mathbb{1}\{\hat{a}_i = a_i\}, \tag{6}$$

where $\mathbb{1}\{\cdot\}$ is an indicator function.

Definition 2. ***(Action Confidence)*** *is defined as the average posterior probability of the selected action $\hat{a}_i$:*

$$\frac{1}{m} \sum_{i=1}^{m} p(\hat{a}_i|\mathbf{x}_i). \tag{7}$$

Definition 3. ***(Action Uncertainty)*** *is defined as the average entropy over the actions in $\mathcal{A} = \{a_j\}_{j=1}^{|\mathcal{A}|}$:*

$$-\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{|\mathcal{A}|} \bar{p}(a_j|\mathbf{x}_i) \log \bar{p}(a_j|\mathbf{x}_i), \tag{8}$$

where $\bar{p}(a_j|\mathbf{x}_i)$ is the probability corresponding to the j-th action in $\mathcal{A}$, and is rescaled and normalized across actions so that $\sum_{j=1}^{|\mathcal{A}|} \bar{p}(a_j|\mathbf{x}_i) = 1$.

In addition to the action-level entropy, we also consider the token-level uncertainty computed over all the tokens in the vocabulary set $\mathcal{V}$:

$$-\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{|\mathcal{V}|} p(v_j|\mathbf{x}_i) \log p(v_j|\mathbf{x}_i), \tag{9}$$

where v_j corresponds to the j-th token in $\mathcal{V}$.

Definition 4. ***(Degree of Alteration)*** *is defined as the Kullback-Leibler divergence between the output probability $p(\cdot|\cdot)$ after persona elicitation and the base model's output probability $q(\cdot|\cdot)$:*

$$-\frac{1}{m} \sum_{i=1}^{m} \sum_{j=1}^{|\mathcal{V}|} p(v_j|\mathbf{x}_j) \log \frac{p(v_j|\mathbf{x}_j)}{q(v_j|\mathbf{x}_j)}. \tag{10}$$

4. Experiments

4.1. Settings

Dataset. For evaluation, we leverage Anthropic's Persona dataset (Perez et al., 2022), which encompasses diverse types of personas. We use the Huggingface version which consists of 99 different personas[1], each entailing 500 statements that align and 500 statements that disagree with the persona trait. We split these 1,000 statements into 700 train samples and 300 test samples randomly, while preserving the label proportion. The train set is used as the sampling pool for ICL baselines and as the training data for our persona fine-tuning phase. The test set is preserved only for evaluation. The list of 99 personas, dataset split indices, and other dataset details are further provided in Appendix A.

Models. We comprehensively evaluate PICLe on three different LLMs: **Llama-2** (Touvron et al., 2023), **Vicuna** (Chiang et al., 2023), and **GPT-J** (Wang, 2021). Specifically, we use the 'llama-2-7b-chat-hf' version for Llama-2, which is a model aligned to human preferences using RLHF. For Vicuna, we use the 'vicuna-7b-v1.5' version, which is a model fine-tuned from Llama-2-base without RLHF. Finally, we utilize the 'gpt-j-6b' version for GPT-J.

Implementation Details. Persona SFT is performed via a next-token prediction objective, wherein every input sample is a fusion of three statements selected from the persona statement pool $\mathcal{S}_{\tilde{\phi}}$. For Persona SFT, we use LoRA (Hu et al., 2021) with rank $r = 8$ and $\alpha = 32$, and train for 4 epochs. For the number of in-context examples, we default to $K = 3$ and perform ablations by varying K in Section 5.1. For the ICL phase, we map semantically equivalent outputs to an action in the binary action set, $\{\text{yes}, \text{no}\}$. Moreover, to encourage the model to respond in a desired way, we also append a system prompt "Answer with Yes or No only" at the end of each input query, for our method and all baselines. Further implementation details are in Appendix B.

[1] https://huggingface.co/datasets/Anthropic/model-written-evals

Table 1. **Persona elicitation results.** Three context examples were used for the ICL baselines. '(Action) Consistency' and 'Action Confidence (Conf)' are in percentage. 'Uncert' refers to the action-level Uncertainty, and 'Tok Uncert' refers to token-level Uncertainty. Best Action Consistency values are in bold.

Methods	Llama-2				Vicuna				GPT-J			
	Consistency↑	Conf	Uncert	Tok Uncert	Consistency↑	Conf	Uncert	Tok Uncert	Consistency↑	Conf	Uncert	Tok Uncert
Base	65.5	95.2	0.1106	0.1199	50.1	62.5	0.3744	0.9181	-	-	-	-
Instructive-prompt	53.9	92.3	0.0905	0.1192	50.0	69.9	0.3821	1.0544	-	-	-	-
Descriptive-prompt	74.9	96.0	0.0616	0.0777	50.4	67.9	0.3362	1.0302	-	-	-	-
Random	79.7	96.0	0.0851	0.0948	72.8	65.4	0.5919	0.9407	58.4	51.8	0.6547	1.7097
Similarity	84.6	96.7	0.0718	0.0793	71.5	64.1	0.6119	0.9459	55.9	51.6	0.6463	1.8251
Uncertainty	69.8	96.0	0.0606	0.0908	76.1	67.6	0.5712	0.9073	59.5	52.1	0.6552	1.6854
Uncertainty-token	80.6	96.0	0.0767	0.0939	73.6	67.5	0.5768	0.8692	54.2	51.5	0.6507	1.7737
Certainty	77.3	95.5	0.1000	0.1069	71.6	63.0	0.6126	1.0147	58.0	51.7	0.6546	1.7304
Certainty-token	71.5	96.2	0.0848	0.0914	71.5	63.2	0.6097	1.0163	61.4	50.7	0.6586	1.7478
Diversity	78.2	95.9	0.0837	0.0956	69.1	65.8	0.5854	0.9578	56.0	52.9	0.6475	1.7100
Likelihood	71.8	95.9	0.0955	0.0992	68.0	61.9	0.6048	1.1006	50.8	53.1	0.6385	1.7914
PICLe (Ours)	**88.1**	97.2	0.0621	0.0679	**78.6**	68.3	0.5744	0.8670	**67.0**	50.9	0.6547	1.7697

4.2. Baselines

4.2.1. Non-ICL baselines

Base is the most basic approach that directly queries the model with a prompt form, "Is the following statement something you would say? `[STATEMENT]`."

Instructive prompting explicitly instructs the model to adhere to the persona by prompting as "Assume that you have or agree with the persona called `[PERSONA]`. Is the following statement something you would say? `[STATEMENT]`."

Descriptive prompting improves the instructive prompting approach by providing a short self-generated description of the target persona. The algorithm for this baseline is: (1) Generate a description of persona with "How would you describe a persona called `[PERSONA]` in one sentence?". (2) Use the generated description as "The persona called `[PERSONA]` can be described as: `[DESCRIPTION]`. Now assume that you have or agree with this persona. Is the following statement something you would say? `[STATEMENT]`." See Appendix C for sample descriptions.

4.2.2. ICL baselines

Random selects K ICL examples randomly from pool $\mathcal{S}_{\tilde{\phi}}$ for each test query $\mathbf{x}_{\text{test}}$.

Similarity selects ICL examples whose sentence embeddings have the highest dot product similarity with respect to the statement in the query. We use the last token embedding extracted from the final layer of the causal language model. See Appendix C for a comparison of the embeddings extracted from different layers.

Uncertainty selects ICL examples with high entropy values defined as $-\sum_{j=1}^{|\mathcal{A}|} \bar{p}(a_j|\mathbf{x}) \log \bar{p}(a_j|\mathbf{x})$, where $a_j \in \mathcal{A}$. The probability is calculated using the prompt "Is the following statement something you would say? `[STATEMENT]`."

We also consider the token-level entropy by computing the entropy of the probability distribution across the entire vocabulary set, similar to our definition in Eq. (9). This approach will be denoted **Uncertainty-token** hereafter.

Certainty is the opposite of "uncertainty" baseline. The ICL examples with the *lowest* entropy values are selected. Similarly, the token-level entropy is considered as well, which will be denoted **Certainty-token** hereafter.

Diversity selects by maximizing diversity. K-means clustering is applied to the sentence embeddings to select K samples that are closest to their respective centroid.

Likelihood selects ICL examples with high log-likelihood values. The log-likelihood of a statement $\mathbf{x}$ is evaluated on the base model as $\log p_\theta(\mathbf{x}) = \sum_{t=1}^{T} \log p_\theta(\mathbf{x}_t|\mathbf{x}_{<t})$, where T is the token length of $\mathbf{x}$.

4.3. Results

Here, we present the main experimental results of Persona Elicitation (Table 1), evalauated on the test datasets.

PICLe consistently outperforms all baselines on three LLMs with respect to Action Consistency. While baselines have no consensus on which approach works best in all three models, our PICLe achieves the highest Action Consistency overall. On Llama-2, PICLe achieves an average action consistency of 88.1%, outperforming the current strongest baseline similarity (84.6%) using the same number of in-context examples ($K = 3$). Moreover, PICLe demonstrates generally high confidence and low uncertainty in its responses, especially when applied to Llama-2. Also see Appendix F for experiment on a bigger Llama-2 model.

PICLe helps non-RLHF models. We verify the performance of PICLe on the non-RLHF models, Vicuna and

Table 2. **Label-aware setting on Llama-2.** Only positive-labeled samples are selected for inference. 3 examples were used for ICL.

Method	Consistency ↑	Confidence	Uncertainty	Tok. Uncert.
Base	65.5	95.2	0.1106	0.1199
Instructive-prompt	53.9	92.3	0.0905	0.1192
Descriptive-prompt	74.9	96.0	0.0616	0.0777
Random	91.5	97.4	0.0611	0.0636
Similarity	92.4	97.7	0.0556	0.0580
Uncertainty	88.6	97.1	0.0686	0.0710
Uncertainty-token	90.5	97.3	0.0644	0.0659
Certainty	92.0	96.9	0.0747	0.0759
Certainty-token	91.6	97.2	0.0680	0.0695
Diversity	92.1	97.6	0.0579	0.0594
Likelihood	87.0	96.9	0.0749	0.0764
PICLe	88.1	97.2	0.0621	0.0679
PICLe $^+$	**93.1**	97.6	0.0577	0.0596

GPT-J. In particular, without ICL, GPT-J completely fails to follow instructions of responding 'yes' or 'no', making it impossible to report any meaningful performances. Vicuna, on the other hand, consistently outputs the same response across different statements, with high confidence. This behavior accounts for Vicuna's Action Consistency of around 50% with near-zero standard deviations. We conjecture that GPT-J and Vicuna being non-RLHF base models contributes to these phenomena. However, when ICL-based methods are applied, these models too show signs of persona elicitation, with significantly increased action consistency values. Notably, PICLe improves the performance from 50.1% (base) to 78.6%, with only three in-context examples.

Refining the selection pool improves ICL performance significantly. In the original experimental settings (Table 1), none of the ICL methods have access to the labels of examples in the pool; they select examples in a label-agnostic manner and persona SFT is done on all persona statements disregarding the labels. Here, we extend the experimental setting to a label-aware setting. Specifically, the ICL baseline methods now select examples from the positive-labeled statements that align with the persona. In Table 2, we observe that this selection pool refinement significantly improves the performance of all ICL methods, when evaluated on Llama-2. For instance, the Action Consistency of the Similarity-based ICL improves from 84.6% to 92.4%. We also demonstrate PICLe$^+$, a variant that only uses the positive-labeled statements for Persona SFT and ICL example selection. The table shows that PICLe$^+$ improves PICLe by 5.0% points, and outperforms the similarity baseline, achieving the best performance overall (93.1%).

5. Analyses

In this section, we provide in-depth analyses of PICLe to better understand its mechanisms and emphasize its advantages. We answer the following research questions:

RQ1. What is the advantage of PICLe and how does it affect model inference? (See section 5.1 and section 5.2)

RQ2. How do different hyperparameter values affect PICLe performance? (See section 5.3)

RQ3. How does the efficiency of PICLe compare with other methods? (See section 5.4)

The analyses are done on Llama-2 unless stated otherwise.

Table 3. **Ablation Study on Llama-2.** 'SFT-likelihood' uses the Persona SFT model likelihood as the score for selection, while 'Original-likelihood' is the "likelihood" baseline in Table 1. Notation $^+$ refers to the label-aware setting as in Table 2.

Method	Consistency ↑	Confidence	Uncertainty	Tok. Uncert.
PICLe	**88.1**	97.2	0.0621	0.0679
SFT-likelihood	72.1	95.6	0.0966	0.1000
Original-likelihood	71.8	95.9	0.0955	0.0992
PICLe $^+$	**93.1**	97.6	0.0577	0.0596
SFT-likelihood $^+$	87.8	96.9	0.0734	0.0750
Original-likelihood $^+$	87.0	96.9	0.0749	0.0764

5.1. Ablation Study

PICLe adopts two models, the original LLM and the persona LLM (*i.e.*, the model after Persona SFT), to compute the log-likelihood difference $\log p_{\tilde{\phi}}(\mathbf{x}) - \log p_{\theta}(\mathbf{x})$ (Eq. (5)). In Table 3, we compare PICLe with baselines that use only one of the two models for likelihood calculation. 'SFT-likelihood' uses only the persona LLM to compute the likelihood of a statement $p_{\tilde{\phi}}(\mathbf{x})$, whereas 'Original-likelihood' is equivalent to the 'likelihood' baseline in Table 1 that uses the original model parameter θ to compute the likelihood $p_{\theta}(\mathbf{x})$. We also study PICLe$^+$, whose SFT model is trained only on positive-labeled statements.

Advantage of using likelihood ratio. The effectiveness of utilizing the likelihood ratio is evident in the results presented in Table 3. PICLe and PICLe$^+$ exhibit a notable advantage over either SFT-likelihood or Original-likelihood. Then, it follows that the advantage comes from the log-likelihood difference we are taking to evaluate a persona statement. This difference quantifies the level of likelihood change required for the Persona LLM to understand a persona statement. A higher degree of change indicates that the statement captures the target persona more thoroughly. Consequently, providing the model with statements that better represent the persona leads to enhanced performance in ICL. Moreover, as the approach prioritizes the relative change in likelihood values, PICLe emphasizes the importance of the shift in distribution rather than the absolute value learned during the Persona SFT phase. Hence, only a single epoch of Persona SFT may be enough to render meaningful differences, making PICLe robust to variations in the number of epochs. Relevant experiment is in section 5.3.